

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

01
INTRODUCTION

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OUTLINE

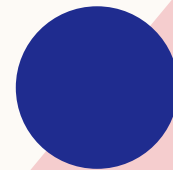
Automata Theory

Symbols

Alphabets

Strings

Languages



AUTOMATA THEORY

- The study of abstract computing devices, or “machines”
- Automata are essential for the study of the limits of computation
 - What can a computer do at all?
 - What can a computer do efficiently?
 - Problems that can in principle be solved, but in practice take so much time that computers are useless for all but very small instances of the problem

WHAT KIND OF THINGS CAN YOU REALLY COMPUTE MECHANICALLY?

- Machine that accepts all binary strings ending with 0

- 10110010

↑ = 0? ✓

= 1? ✗

- Machine that accepts all valid Java codes

- Compilers

↓
binary → valid?
invalid?

- Machine that accepts all valid Java codes and never allows the codes to go into infinite loop?

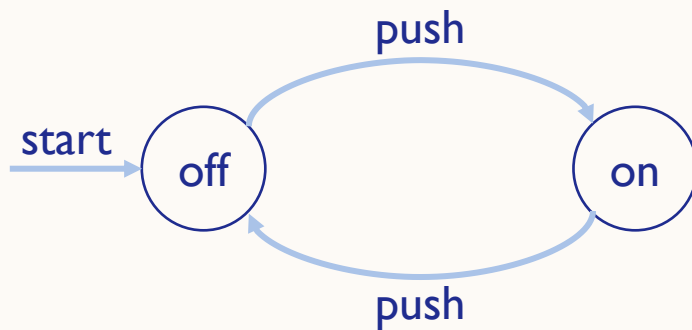
FINITE AUTOMATA

A USEFUL MODEL FOR MANY IMPORTANT KINDS OF HARDWARE AND SOFTWARE

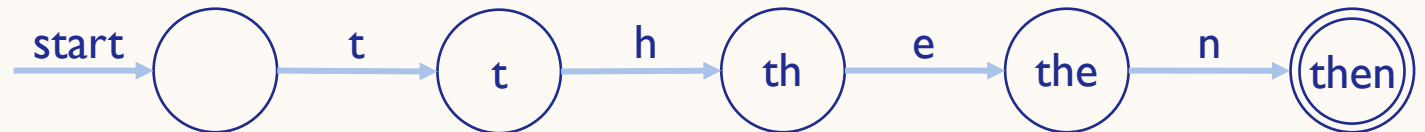
- Software for designing and checking the behavior of digital circuits
- The “lexical analyzer” of a typical compiler
 - The compiler component that breaks the input text into logical units, such as identifiers, keywords, and punctuation
- Software for scanning large bodies of text, such as collections of Web pages, to find occurrences of words, phrases, or other patterns
- Software for verifying systems of all types that have a finite number of distinct states, such as communications protocols or protocols for secure exchange of information

EXAMPLES

A finite automaton modeling an on/off switch



A finite automaton that could be part of a lexical analyzer modeling recognition of then



THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Symbols
 - 0, 1, 2, x, y, z
- Alphabets (Σ)
 - A finite, nonempty **set of symbols**
 - $\Sigma = \{0, 1\}$
 - $\Sigma = \{a, b, \dots, z\}$
- Strings
 - A finite **sequence of symbols** chosen from some alphabet
 - $\Sigma = \{0, 1\}$
 - String: 01101, 111

THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Strings
 - The Empty String (ϵ)
 - The string with zero occurrences of symbols
 - Length of a String ($|w|$)
 - The number of positions for symbols in the string
 - 01101 has length 5
 - $|011| = 3$
 - $|\epsilon| = 0$

THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Strings
 - Powers of Σ
 - $\Sigma = \{0, 1\}$
 - Σ^0 = set of all strings of length 0 = $\{\epsilon\}$
 - Σ^1 = set of all strings of length 1 = $\{0, 1\}$
 - Σ^2 = set of all strings of length 2 = $\{00, 01, 10, 11\}$
 - Σ^3 = set of all strings of length 3 = $\{000, 001, 010, 011, 100, 101, 110, 111\}$
 - Σ^k = set of all strings of length k
 - Cardinality
 - Number of elements in a set
 - $\Sigma = \{0, 1\}$
 - $\Sigma^k \rightarrow \text{cardinality} = 2^k$

THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Strings

- $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$

- The set of all strings over an alphabet Σ

- $\Sigma = \{0, 1\}$

- $\Sigma^* = \{\epsilon\} \cup \{0, 1\} \cup \{00, 01, 10, 11\} \cup \dots = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, \dots\}$

- Set of all possible strings of all lengths over $\{0, 1\}$

infinite

- $\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$

- The set of nonempty strings from alphabet Σ

- Exclude the empty string from the set of strings

- $\Sigma^* = \Sigma^+ \cup \{\epsilon\}$

THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Strings
 - Concatenation of Strings
 - x is the string composed of i symbols
 - $x = a_1a_2\dots a_i$
 - y is the string composed of j symbols
 - $y = b_1b_2\dots b_j$
 - xy is the string of length $i + j$
 - $xy = a_1a_2\dots a_ib_1b_2\dots b_j$
 - $x = 01101$ and $y = 110$
 - $xy = 01101110$ and $yx = 11001101$
 - $\epsilon W = W\epsilon = W$

THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Languages
 - A **set of strings** all of which are chosen from some Σ^*
 - English
 - The collection of legal English words is a set of strings over the alphabet that consists of all the letters
 - C programming language
 - The legal programs are a subset of the possible strings that can be formed from the alphabet of the language (a subset of the ASCII characters)
 - $\Sigma = \{0, 1\}$
 - $L_1 = \text{set of all strings of length 2} = \{00, 01, 10, 11\}$
 - $L_2 = \text{set of all strings of length 3} = \{000, 001, 010, 011, 100, 101, 110, 111\}$ **finite**
 - $L_3 = \text{set of all strings that begin with 0} = \{0, 00, 01, 000, 001, 010, 011, 0000, \dots\}$ **infinite**

THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Languages
 - The language of all strings consisting of n 0's followed by n 1's, for some $n \geq 0$
 - $\{\epsilon, 01, 0011, 000111, \dots\}$
 - The set of strings of 0's and 1's with an equal number of each
 - $\{\epsilon, 01, 10, 0011, 0101, 1001, \dots\}$
 - The set of binary numbers whose value is a prime
 - $\{10, 11, 101, 111, 1011, \dots\}$

THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Languages
 - The empty language
 - $\emptyset$
 - Has no strings
 - The language consisting of only the empty string
 - $\{\epsilon\}$
 - Has one string

THE CENTRAL CONCEPTS OF AUTOMATA THEORY

- Languages
 - Set-Formers as a Way to Define Languages
 - $\{w \mid \text{something about } w\}$
 - The set of words w such that (whatever is said about w to the right of the vertical bar)
 - $\{w \mid w \text{ consists of an equal number of 0's and 1's}\}$
 - $\{w \mid w \text{ is a binary integer that is prime}\}$
 - It is also common to replace w by some expression with parameters and describe the strings in the language by stating conditions on the parameters
 - $\{0^n 1^n \mid n \geq 1\}$
 - The set of 0 to the n 1 to the n such that n is greater than or equal to 1
 - $\{01, 0011, 000111, \dots\}$
 - $\{0^i 1^j \mid 0 \leq i \leq j\}$
 - Strings with some 0's (possibly none) followed by at least as many 1's



PRACTICE

EXERCISES

1. $|\text{babba}| = \dots$
2. $|\epsilon| = \dots$
3. $\Sigma = \{0, 1, 2\} \rightarrow \Sigma^2 = \dots$
4. $\Sigma = \{p, q, r, s\} \rightarrow \text{cardinality of } \Sigma^3 = \dots$
5. $\Sigma = \{a, b, c\} \rightarrow \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 = \dots$
6. $p = 10110$ and $q = 0101 \rightarrow pq = \dots$
7. $L = \{\text{set of all strings over } \{0, 1\} \text{ of length } 5 \text{ that end with '10'}\} = \dots$

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

NEXT

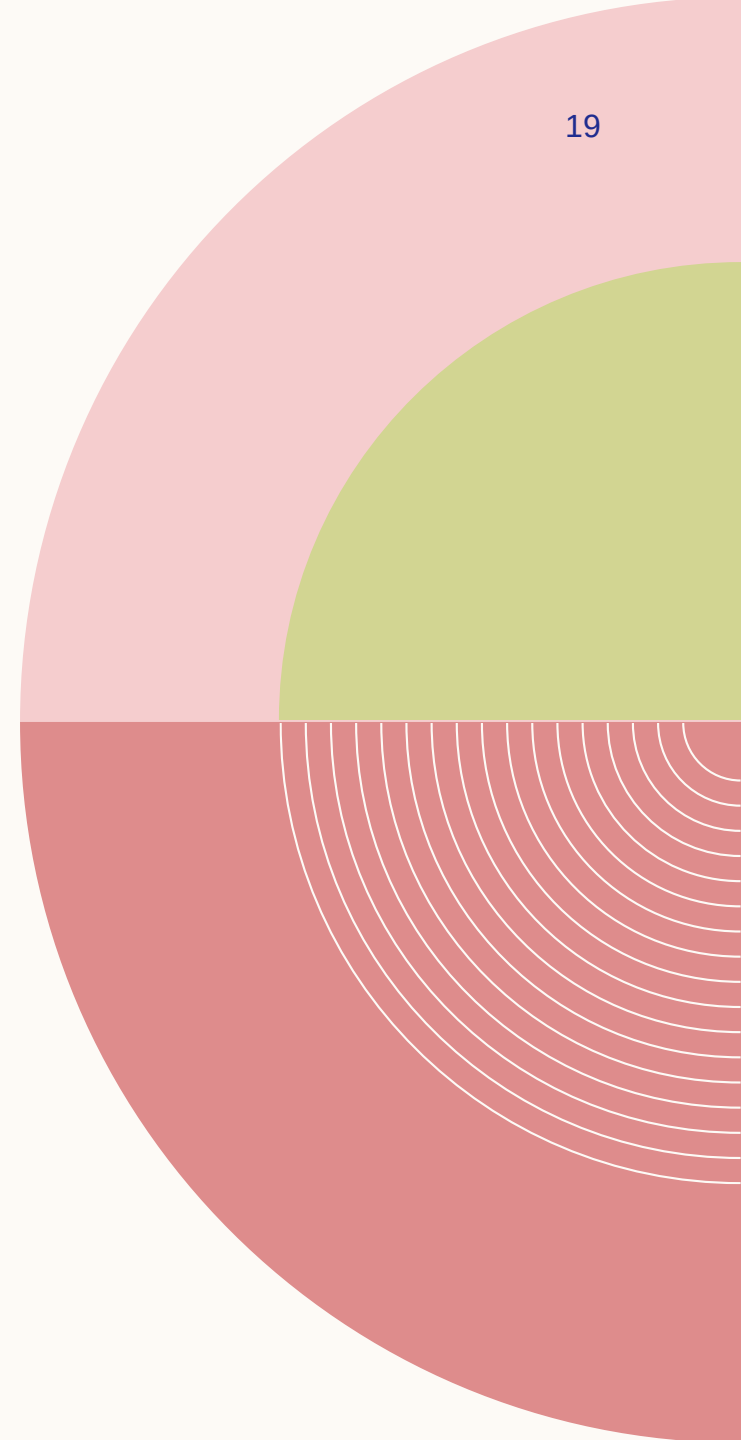
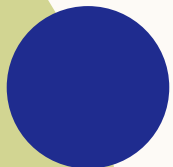
Deterministic Finite Automata (DFA):

Finite State Automata (FSA)

Deterministic Finite Automata (DFA)

Regular Languages

Equivalence of DFA



VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



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